MIL-P-81237(WP)
5 March 1965

SUPERSEDING
WS 1711
6 November 1962

MILITARY SPECIFICATION
PENTAERYTHRITOL DIOLEATE

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers one type of pentaerythritol dioleate for use as an ingredient of a wetting agent.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

(When requesting any of the above documents, give the title and complete designation of the item shown above. Copies of this specification and other unclassified specifications, standards, and publications required by contractors in connection with specific procurement functions may be obtained from the Commanding Officer, Navy Supply Depot (CDS), 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

PUBLICATIONS

American Society for Testing and Materials

ASTM D 1544-58T	Color of Transparent Liquids (Gardner Color Scale)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103.)

3. REQUIREMENTS

3.1 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the pentaerythritol dioleate shall be manufactured using the methods and procedures proposed for the production. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract. After satisfactorily passing all the preproduction tests specified herein, no changes in raw materials and processing of materials for production shall be made without prior written approval of the procuring activity.

3.2 Data requirements. No data is required by this specification or by applicable documents referenced in Section 2 unless specified in the contract or order (see 6.2).

3.3 Physical and chemical properties

3.3.1 Saponification number. The saponification number of the pentaerythritol dioleate shall be 169 ± 9.

3.3.2 Color. The Gardner Scale color of the pentaerythritol dioleate shall be a maximum of 14.

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3.3.3 Acid number. The acid number of the pentaerythritol dioleate shall be a maximum of 7.5.

3.4 Workmanship. The material shall be uniform in quality and shall be free from impurities and other defects that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the pentaerythritol dioleate shall be classified as follows:

a. Preproduction inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Sampling

4.3.1 Preproduction sample. After mixing until the material is homogeneous, a preproduction sample of four, 8-ounce samples shall be removed from each container. Combine any two of the four samples to form a composite sample. Repeat with the remaining two samples to form the second composite sample. Each composite sample manufactured in accordance with 3.1 shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Further production of the pentaerythritol dioleate by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Quality conformance inspection sampling. Unless otherwise specified, sufficient material shall be taken from each lot to perform the tests as specified in 4.5.

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4.3.3 Inspection lot. An inspection lot of pentaerythritol dioleate shall consist of all material presented for acceptance at one time and produced in a single manufacturing batch under homogeneous conditions of manufacture.

4.4 Preproduction inspection. The preproduction sample shall satisfactorily pass the quality conformance inspections detailed in 4.5.

4.5 Quality conformance inspection

4.5.1 Visual examination. Visually examine all containers in the lot to determine compliance with 3.4 and Section 5.

4.5.2 Saponification number test. Transfer approximately 3.5 grams weighed to the nearest 0.1 milligram (mg) of the sample into a 300 milliliter (ml) ground-glass Erlenmeyer or round-bottom flask. Add 50 ml of approximately 0.5 normal (N) alcoholic potassium hydroxide solution. Connect the flask to an air condenser, approximately 250 millimeter (mm) in length, and reflux for 3 hours on a steam bath at such a rate that the major portion of the alcohol will condense in the lower half of the condenser. Allow the flask and contents to cool to room temperature. Remove the condenser and rinse with three 10-ml portions of neutralized 95-percent alcohol, collecting the washings in the flask. Add 2 or 3 drops of 1-percent phenolphthalein indicator to the flask contents, and titrate with standardized 0.5N hydrochloric acid to the appearance of colorless solution. Make a blank determination on 50 ml of the alcoholic potassium hydroxide solution in accordance with this procedure. Calculate the saponification number as follows:

$$\text{Saponification number} = \frac{56.1\ (C-B)\ N}{W}$$

where:

B = ml of hydrochloric acid required to titrate the sample
C = ml of hydrochloric acid required for blank titration
N = Normality of hydrochloric acid solution
W = Weight of sample, grams

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4.5.3 Color. Prepare a set of Gardner color standards according to ASTM D 1544-58T. Fill a Gardner sample tube and compare the color of the sample to that of the standard. Record the number of the standard that most closely matches the sample as the color of the sample.

4.5.4 Acid number. To 150 ml of absolute ethanol in a 250-ml flask add 2 drops of 1-percent phenolphthalein indicator. Titrate with 0.1 N sodium hydroxide to a faint pink end point that persists for at least 30 seconds. Add about 10 grams of composite sample, weighed to the nearest 0.1 mg, and stir to dissolve, warming slightly if necessary. Titrate the sample solution with 0.1 N sodium hydroxide to the faint pink end point that persists for at least 30 seconds. Calculate the acid number as follows:

$$\text{Acid number} = \frac{56.1 \times V \times N}{W}$$

where:

V = Volume of sodium hydroxide to titrate sample solution, ml
N = Normality of sodium hydroxide
W = Weight of sample, grams

4.6 Acceptance criteria. Failure of any preproduction or quality conformance sample to comply with any requirement of this specification shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified in the contract or order, minimum requirements for preservation, packaging and packing shall be in accordance with MIL-P-116, Method III.

5.2 Marking. Each container shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

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a. Manufacturer's name and location
b. Material trade name
c. Net weight or volume
d. Lot number, batch number and date of manufacture
e. Shelf life or storage limitations
f. Number and revision letter of this specification

6. NOTES

6.1 Intended use. The material purchased in accordance with this specification is intended to be used as an ingredient in the chemical composition of a wetting agent.

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following information:

a. Title, number and revision letter of this specification
b. Minimum lot size, if applicable
c. Place of delivery
d. Size of container
e. Request for test data
f. Whether preproduction sample is required

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-P-81237(WP) PENTAERYTHRITOL DIOLEATE

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Bureau of Naval Weapons
Washington, D. C. 20360

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